WebToSpeech

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Abstract

WebToSpeech: a web application designed to provide a simplified, organized reading experience for a wide audience. Users will be able to upload their documents to a database and have them read aloud using text-to-speech synthesis. Organizing a large collection of books and documents can be tedious and disrupt the reading process. To remedy this, WebToSpeech also provides a central hub to organize documents efficiently, sorting them into collections made by the user. Users will be able to create collections by name, add files to those collections, and filter the displayed documents to find only the ones they are searching for. Finally, users will have the option to add annotations and bookmarks to their documents. This will improve the user experience for online document viewing and provide accessibility services to those with reading disabilities.

Use Cases

WebToSpeech will have several use cases. WebToSpeech can provide accessibility options for users with hearing impairment issues. Individuals who suffer from dyslexia and other similar issues can utilize WebToSpeech to make reading easier for them. Users can use the site to play documents in the background while focusing on another task, thereby increasing their productivity.

Technology

WebToSpeech utilizes several technologies and APIs that can be found online. The website was designed using React. User accounts and data storage are handled by Supabase. The website is hosted online using Vercel. The website utilizes the Web Speech API to handle the text-to-speech functions of the website.

Design

WebToSpeech features 4 primary pages: the sign-in page, the dashboard page, the document page, and the profile page. The sign-in page will be a very simple page that will prompt the user to sign in with Google. This will be the only way available to sign in and will be handled using Supabase. The dashboard page will serve as a home page for the website. On this page, the user will be able to upload documents as well as view documents previously uploaded. The document page is where the user can view the contents of uploaded documents and have the document read aloud to them. This page features controls and annotation options at the bottom of the page. The profile page is where the user can see information about their profile. The number of documents a user has will be displayed here. The user may also adjust their account settings on this page.

Figure 1: A mobile view of the profile page of the website

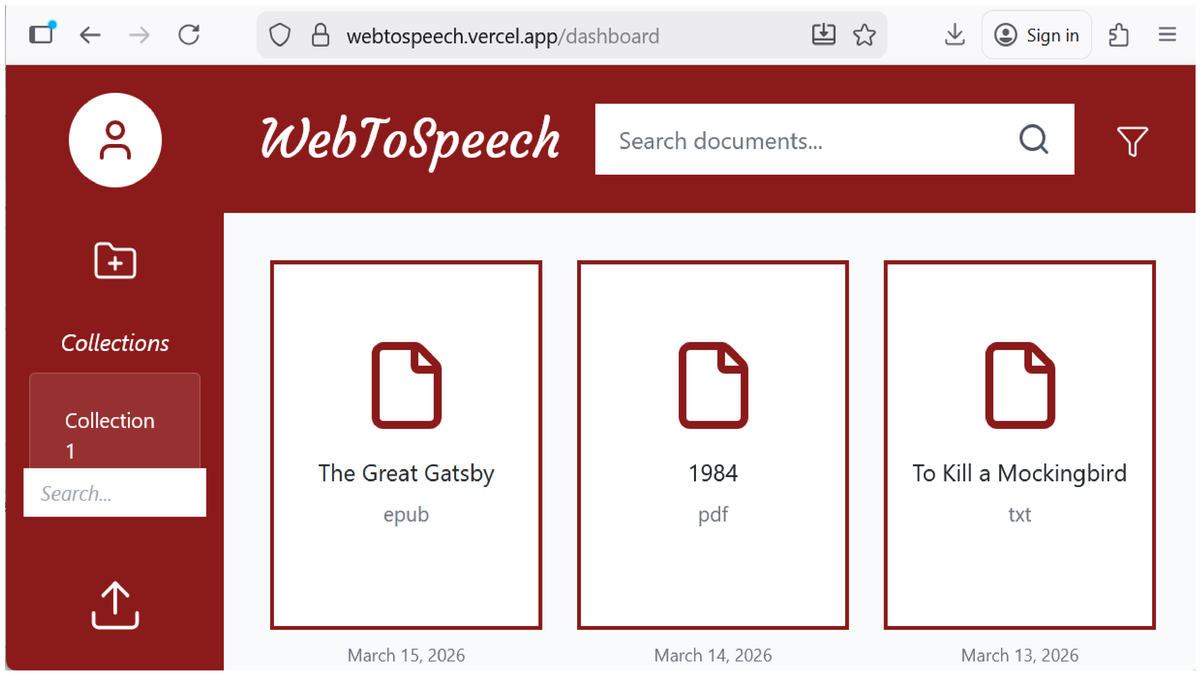

Figure 2: A desktop view of the dashboard page

Planned Features

If this project were to continue forward, the following features are planned to be added to the website.

1. A way to share profiles and collections with others
2. More advanced annotation controls in the document section
3. More profile options available for users
4. More file types supported for uploads

References

1. React: https://react.dev/.
2. Vercel: https://vercel.com
3. Supabase: https://supabase.com
4. API: Web Speech API

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